Mining Software Repositories

Chapter 8: Validity of Results

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Aspects of validity

Trustworthyness of results

- Imagine you study the claim that that code based on pattern matching is easier to read and leads to less bugs than code using traditional if/else or switch statements

```java
if (shape instanceof Rectangle r) {
    return 2 * r.length() + 2 * r.width();
} else if (shape instanceof Circle c) {
    return 2 * c.radius() * Math.PI;
} else {
    throw new IllegalArgumentException("Unrecognized shape");
}
```

```java
return switch (shape) {
    case Rectangle r -> 2 * r.length() + 2 * r.width();
    case Circle c    -> 2 * c.radius() * Math.PI;
    default          -> throw new IllegalArgumentException("Unrecognized shape");
}
```

https://docs.oracle.com/en/java/javase/17/language/pattern-matching.html

- Would you trust results based on …
  - … asking two developers? 20? 100?
  - … showing two code snippet? 20? 100?
  - … the complexity of the examples?
  - … asking first year students, senior students, junior developers, or senior developers?
  - …

Validity of results

- In science, trust is not based on a feeling, but rather on the validity of results
- Consequently, we need to understand how our methods and data affect the validity of our results with respect to our research question
- Some aspects of validity were already discussed
  - The representativeness of samples
  - Why and how to use statistics
  - Coding data with multiple raters and measuring their agreement
- Within this chapter, we look at validity in a more structured manner
  - We look at the treats to validity, i.e.., aspects that could reduce validity
  - Note that a threat does not invalidate (or even falsify) results. It just makes them less trustworthy.

Breaking down validity

- There are many aspects that affect the validity of our results and also mutliple perspectives on the validity
- A commonly used approach is based on the work from Cook et al. and popularized in software engineering by Wohlin et al.
- Here, we consider validity in five dimensions
  - Construct validity to consider if what the study measures or analyzes is representative for the concepts that are studied
  - Internal validity to consider if what we observe is supports causal claims, i.e., that we we observe is really causaly related to the effects we want to study
  - External validity to consider how the results are generalizable beyond the scope of the study
  - (Statistical) Conclusion validity to consider if the statistical methods used likely yield valid results
    - Only used in case research methods use statistics
  - Reliability to consider if there may be biases introduced by the researchers and if different researchers would reach the same conclusions
    - Only used in case of qualitative research methods

Cook, T. D. (1979). Quasi-experimentation: Design and analysis issues for field settings. *Rand McNaly College Publishing*.

Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). *Experimentation in software engineering* (Vol. 236). Berlin: Springer.

Construct validity

Design threats to the construct validity(1/2)

- Design threats consider how the study design/methods affect the validity of the construct

- Inadequate pre-operational explication of constructs
  - The theory, hypothesis, or concept that is studied is not sufficiently clear, which means that we cannot really know if we measure what we want to
  - Example: „A good process yields high-quality software“ – neither good process nor high-quality software is sufficiently detailed
- Mono-operation bias
  - If we only consider a single independent variable, case, subject, or similar, there is a chance that we do not get the full picture
  - Example: Using a single project to study a phenomenom
- Mono-method bias
  - A single perspective on a concept (even when measured with mutliple variables) has the risk of missing other imporant aspects
  - Example: Studying developer communication only using the issue tracker, but not other communication media

Design threats to the construct validity(2/2)

- Confounding construct
  - Factors that could also explain observed effects are not considered
  - Example: Studying readbility without controling for general prior programming skills and skills in considered programming language
- Interaction of different treatments
  - When multiple treatments are applied at the same time, they may interact
  - Example: When a project uses introduces multiple new quality assurance tools at the same time, it is hard to measure how each tool contributed to changes in software quality
- Interaction of testing and treatment
  - The study itself may affect the outcomes
  - Example: When studying how developers use a certain tool, it makes a differences if they know whether they are studied or not
- Restricted generalizability across constructs
  - When a specific aspect is studied there may be side-effects to related constructs which may hinder generalization of positive effects
  - Example: Writing code with the help of an AI may affect the speed at which new features are added, but it needs to be controlled how this affects the maintainability to understand if overall development speed is positively of negatively affects

Social threats to construct validity

- Social threats are issues that may occur even if the design is valid

- Hypothesis guessing
  - Participants in an experiment might guess what a study is about and alter their behavior to affect the results
  - Example: When judging readability, personal favor of functional/procedural might affect judgment if participants guess that this is what the study is about
- Evaluation apprehension
  - Humans may be afraid to be evaluated and behave differently, e.g., to “look better“ or to “meet expectations“
  - Example: When judging if something is a bug, raters may be overly criticial for fear of looking bad because they missed something
- Experimenter expectancy:
  - Similar to hypothesis guesses, researchers might also consciously or unconsiously bias results based on their expectations
  - Example: If we expect that code review is low quality, we may be overly critical when judging the process

Internal validity

Single group threats to internal validity (1/3)

- Such threats apply if we do not have a control group
  - In the treatment group, we study some phenomonem that takes place
  - The control group is a second group of data, where this does not happen
  - Thus, such threats consider how we misjudge what we find if we only study what happens under a treatment without checking what happens without this
  - Example: We study projects using code review as treatment group and projects without code review as control group. In this case, code review is the treatment.

- History
  - If we apply multiple treatments over time, when this takes place might affect the results
  - Example: When studying developer productiviy over intervals of one week, it might matter if this immediately before or after holidays
- Maturation
  - Treatments may have a different effect over time
  - Example: their might be initial skepsis or euphorie for a new technology and results measured at a later point in time might be different

Single group threats to internal validity (2/3)

- Testing
  - Data might change if the test subjects know that a test is occuring, especially if this is repeated
  - Example: Developers might alter how they write code to affect to have a higher average cyclomatic complexity if they know that they are the difficulty of problems they worked is judged based on this metric. (note: related to social biases)
- Instrumentation
  - If we need to modify our measurement object for the measurement, this instrumentation might have an effect
  - Example: If we modify a CI system to report certain results over time for a mining study, this might affect other aspects of the CI system, e.g., it could be updated more often than usual or have more bugs
- Statistical regression:
  - When we select data based on prior results, this could affect what is possible to observe
  - Example: If we select the top-10 performers from the past year, an improvement is not likely as these were already top performers. On the other hand, if we select bottom-10 performers, getting worse is also not likely.
- Selection:
  - How samples are selected may affect how representative they are
  - Example: see sampling in methods chapter!

Single group threats to internal validity (3/3)

- Mortality
  - Study subjects might drop out during a study making the final sample less representative than initially desired and achieved
  - Example: When studying longitudinal effects of tools on software projects, the activity of some projects might decrease/stop

- Ambiguity about direction of causal influence
  - Whenever we study correlations we need to ensure direction of causality when we want to make such claims
  - Example: When studying if complex code causes more defects, a possible alternative explanation is that both complex code and defects are caused by the problem complexity. Thus, there would be a correlation, but the cause would not be the code complexity, but the problem complexity. (note: related to confounding factors!)

Multiple group threats to internal validity

- Control groups are often using within study designs to study to mitigate internal threats

- However, using multiple groups (e.g., control group, treatment group, different treatments) can also lead to validity problems

- Interaction with selection
  - Any of the above threats might interaction with the group selection
  - Example: We might select a control group of projects without code review. However, this group might mature faster, leading to a decreasing defect rate, thereby confusing results.

Social threats to internal validity

- There can also be social aspects that influence what we observe. This is often somewhat related to the construct.

- Diffusion or imitation of treatment
  - One group might learn about a treatment of another group and try to imitate this because they suppose there is a positive effect
  - Example: You want to study how reporting of static analysis warnings affects the number of bugs. One project has such reporting, the other does not. Developers from the second project might then start to use the static analysis locally more often to imitate this.
- Compensatory rivalry
  - A group that gets a less desirable treatment might be motivated by this and affect the outcome
  - Example: If one group gets to use AI support, the non-AI group might be motivated to show that they can outperform development with AI support
- Resentful demoralization
  - Opposite of compensatory rivalry

External validity

Threats to the external validity

- Threats to external validity consider if we can apply our results in context's different from the study and thus consider how the context can vary

- Interaction of selection and treatment
  - Such threats occur when the selected sample is not representative for the population we want to generalize too
  - Example: It is unclear how MSR studies on open source software generalize to proprietary, closed-source, industrial projects
- Interaction of setting and treatment
  - Such threats occor then the setting in which the study is conducted may have an impact on the observed effects
  - Example: When we study the impact of code-review in a company that uses a modern inner-source tool environment this may be different from more date tooling that has more friction.
- Interaction of history and treatment
  - Such threats occur when the timing of the experiment might affect the general representativeness, similar to the internal threat "history"
  - Example: Measuring average developer activity in August is likely not represenative for regions that currently have their summer holiday season

Conclusion validity

Threats to conclusion validity (1/3)

- Threats to the conclusion validity arise from the probabilisitic nature of statistics
- Thus, they consider aspects that lead increase or lead to a high probability of (meaningful) errors in the results

- Low statistical power
  - Low statistical power means that the likelihood of accepting the null hypothesis is high
  - Example: Typically the case when the sample size is too small or the effect size is small

- Violated assumptions of statistical tests
  - Assumptions on the distribution of data are not completely fulfilled
  - Example: Data for an attribute looks almost normal, so a Cohen‘s d is used due to the high interpretability, even though the data is not completely normal.

Threats to conclusion validity (2/3)

- Fishing and error rate
  - Fishing is when we continue to look at different aspects, until we find a result that supports our hypothesis. This often also results in an inadvertantly increased error rate due to repeated tests.
  - Example: We want to know if complexity has a relationship to post-release defects. However, we use twenty different, but related complexity measures, increasing the chance that we randomly find results supporting the hypothesis.
- Reliability of measures
  - If we have measurements that have a high uncertainty, this affects downstream statistical results
  - Example: We use a Likert scale to let participants judge the readibility of mined code. If the results on the Likert scale are unstable, this affects the statistical analysis.
- Reliability of treatment implementation
  - If we have a treatment that does not always yield the same outcome, this could affect our statistical results
  - Example: When using AI support, developers may use this differently, e.g., to generate database queries or to implement requirements. These aspects may work with different accuracies and the distribution how the tasks are selected affects the outcome of the statistics.

Threats to conclusion validity (3/3)

- Random irrelevancies in the experimental setting
  - Results can be affected by external effects that randomly occur
  - Example: Any MSR measurement in April 2020 will have an effect due to Covid-19

- Random heterogeneity of subjects
  - We may be unlucky with our sampling due to the natural heterogeneity within groups
  - Example: If I randomly pick 20 students, there is a chance that they are not representative for the whole student population, because the overall population is so diverse

Reliability

Threats to the reliability

- Threats to the reliability are the result of methods that rely on the expertise of researchers, but do not have safeguards or check to prevent biases from these researchers to affect the data

- Insufficient details about data collection
  - Details about the data collection are missing such that a replication might collect different data leading to a different sample strategy or difference in how data is augmented and enriched
  - Example: An open coding of code comment types does not describe if and how codes are harmonized between raters
- Unclear survey questions
  - Survey questions are not clearly worded and require interpretation by the researcher leading to different formulations
  - Example: The question “How would you judge complexity?“ could refer to code or other aspects of software development, and within code this could be on different levels of granularity, e.g., file level, class level, or function/method level.
- Unreliable judgements
  - Raters might have biases or judgements may be difficult, influencing the reliability of the data
  - Example: A single researcher is asked to analyze which kinds of components (e.g., UI, database, networking) are affected by issues. Due to the difficulty of this task it is likely that there are some mislabels

Reporting threats to validity

No study is perfect

- Time and effort we can invest is limited
- Some aspects for validity are actually in a trade-off relationship
  - E.g., you can increse internal validity by looking at more possible confounding factors or using a more in-depth analysis method for each data point to establish causality
  - However, given a fixed budget this effort is at the cost of external validity as fewer data points can be considered, affecting the generalizability
- Studies can and should be open about threats to validity
  - Identify relevant aspects based on the lists we have above
    - Note that not everything is relevant for every study
  - Judge how strong the threat is and what it might mean for the results
    - This needs to be done in context of the results!
    - E.g., if you have a large effect size on a small sample, it is likely that there is an effect with a larger sample, though it might not be as strong. Thus, the threat is rather towards the strength of the effect, not the effect itself
    - However, if you have a small effect size and a small sample, the threat is directly to the effect which may be a false positive

Example for reporting of threats Construct validity

- First repeat the aspect of the method that leads to a threat
  - "We differentiate not only between bug fixing lines and other lines, but use a more detailed differentiation of changes into whitespaces only, documentation changes, test changes, refactorings, and unrelated improvements."
- Then state what the impact of the threat could be
  - "This could have influenced our results. Most notably, this could have inflated the number of lines without consensus.
- Then state if there are any mitigating aspects in your method or how you took this into account when interpreting data
  - "We counter this threat by differentiating between lines without consensus with at least one bug fix label, and other lines without consensus."
- If there are no mitigating aspects, you need to instead state how you would expect results to possibly change
- You can possibly also state why a mitigate was not possible or not used
  - E.g., due to effort or because this would lead to other problems

Herbold, S., Trautsch, A., Ledel, B. *et al.* A fine-grained data set and analysis of tangling in bug fixing commits. *Empir Software Eng* **27**, 125 (2022). https://doi.org/10.1007/s10664-021-10083-5

Examples for reporting threats Internal validity

- Same general structure: Threat (red), mitigation or analysis of threat (green)
  - "Our decision to label data on a line level and not the character or statement level may affect the validity of our conclusions. In case changes are tangled within the same line, this is not detected by our labeling, since we advised participants to label lines as bug fixing in that case. However, such changes are very rare, as this requires not only lines with multiple concerns, but also that these different concerns are modified within the same commit, such that at least one modification is not part of the bug fix. While we do not have data on how often exactly that was the case, our participants reported that this happened only rarely. Thus, any noise we missed this way is likely to have only a negligible effect on our results."

- Diffentiation between construct and internal validity often difficult
  - In general, it if rather regarding what you measure, it is construct validity
  - If it is rather how you interpret your findings, it is internal validity

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Examples for reporting threats External validity

- Reporting of external validity is almost always done with respect to the sample
  - Typically, this is rather conservative, i.e., restricts the scope of where findings can be applied to the immediate population of the sampling frame
  - Broader generalizations are possible, but often critizised by reviewers

- Discussion of such threats should refer to the representativeness of the sample and argue based on properties of the sampling approach
  - "Due to our restriction to mature Java software, our results may not generalize to other languages or software with less mature development processes than those of the Apache projects. Moreover, the sample of projects may not be representative for mature Java projects as well, as we did not conduct random sampling, but re-used data from Herbold et al. (2019) who used a purposive sampling approach (Patton 2014). However, Baltes and Ralph (2020) argue that well-done purposive sampling should also yield generalizable results. Moreover, our analysis was done on open-source projects. While there are many industrial contributors in the projects we studied, our results may not generalize to closed-source software."

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Examples for conclusion validity

- Ideally, your study is fully sound, i.e., has sufficient data and all assumptions for your tests are fulfilled
  - "There are no threats to the conclusion validity of our study. All statistical tests are suitable for our data, we correct our results to err on the side of caution, and the pre-registration ensured that we did not tailor our statistical analysis in a way to artificially find coincidental results, e.g., through subgroup analysis."

- However, often you have some aspects that may not be perfect
  - "We report the effect size using Cohen's *d*. While our data is not perfectly normal, it also does not have severe outliers, so we prefer the clear interpretation of Cohen's *d* over the slightly more appropriate, but less accessible non-parametric effect size measures."

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Examples for reporting threats Reliability

- Threats to the reliability are reported in a similar manner to threats to the construct
- Ideally, we can argue why the results should be reliable
  - "We cannot rule out that the results of our study are due to the participants we recruited as fellow researchers. However, our participants are very diverse: they are working on all continents except Africa and Antarctica, the experience ranges from graduate students to full professors, and from a couple of years of development to several decades. Regardless of this, the overall consensus ratios are relatively high for all participants and Fleiss' κ indicates substantial agreement among our participants."
- However, even if results are not fully reliable, usually seen through low agreement between researchers, this does not automatically mean that they are invalid
  - "Due to the subjective nature of the interpretation of the quality of an explanation, there is a non-negligible threat that the study results could be different, if the study were to be executed by a different set of researchers. As we discuss in Section 6.6, it seems like **this threat is more relevant to individual data points, rather than overall trends**. Thus, while we believe that especially the coefficients are uncertain, our data points in the direction of a reliable conclusions based on the general trends. Nevertheless, we believe that it would be extremely interesting to see an exact replication of ourwork to better understand the reliability issues are in work here, as they may not only be related to this study, but to the assessment of XAI approaches in general."

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Threats to validity vs limitations

- Limitations of a study are often mixed up or seems as the same as the threats to the validity
  - From a certain perspective, this makes sense: threats to validity *limit* the trust in the study results
- However, limitations are rather regarding things that are important but not yet considered
  - Could be other populations
  - Could be other perspectives on the problem
  - Could be other, related research questions or other aspects of a theory that were not yet studied sufficiently
- Threats on the other hand are only for the scope of the current study and do not go beyond this scope
  - Notably, nothing that is out of scope of the study (e.g., other research questions) can be a threat to the validity
- However, there is certainly an overlap
  - E.g., other populations that need to be studied are directly related to the external validity

Questions?

